Rethinking Remote Sensing Pretrained Model: Instance-Aware Visual Prompting for Remote Sensing Scene Classification

Leyuan Fang, *Senior Member, IEEE*, Yang Kuang, Qiang Liu, Yi Yang, and Jun Yue

***Abstract*— Large-scale pretrained models, such as vision transformers (ViTs), have made significant progress in remote sensing (RS) scene classification tasks. For a new scene classification task, it is popular to fully fine-tune the pretrained model parameters to avoid training from scratch. Although such an approach achieves satisfactory results, it will lead to heavy computation and storage burden, which limits the transferability of large pretrained models to different RS scene classification tasks. To address this challenge, we propose a parameter-efficient tuning approach called as the instance-aware visual prompting (IVP), which is the first work to explore the prompting in the field of RS scene classification. The proposed IVP adaptively generates prompts based on the complex background and highly variable characteristics of RS images and updates only a few parameters to transfer the pretrained RS transformer model to different scene classification tasks. Specifically, instead of adapting the entire model parameters, we introduce some instance-specific prompt vectors into the input space. Then, considering the significant variability in RS images, we introduce an instance-level prompt generation module to generate specific prompts for each RS image by aggregating contextual information from the input. Finally, these prompt vectors will calibrate the pretrained features to encode instance-specific information. Extensive experiments on three RS scene classification datasets demonstrate the superiority of IVP over other fine-tuning methods. For example, when updating just 1.1% parameters, the Swin transformer (Swin-T) model achieves about 1.83% and 1.42% improvement compared with the full fine-tuning method on NWPU-19 and NWPU-28, respectively.**

***Index Terms*— Image processing, instance-aware prompting, pretrained model, remote sensing (RS) scene classification, transfer learning.**

I. Introduction

With the advancements in Earth observation technology, acquiring high spatial resolution images containing complex and abundant structural information about the Earth's surface has become much easier [1], [2]. However, further interpretation and discrimination of these images is a tremendous challenge. Remote sensing (RS) scene classification is an essential part of interpretation and has garnered significant attention within the RS community. In recent years, scene classification has been widely applied in many fields, such as natural disaster detection [3], geospatial object detection [4], geographic image retrieval [5], [6], environmental monitoring [7], and urban planning [1], making it a crucial area of research.

Initially, researchers directly use simple and intuitive attributes to recognize different aerial scene images, such as color [8], texture [9], contour [10], and spectral information, either alone or in combination [11]. Although these manual features usually perform well in simple scenes, they are less effective in more complex scenes. To further improve the classification performance, some researchers provide a feasible way to automatically extract the image features by learning mapping functions or filters [12]. While these methods are easy to understand and apply, they have not been able to capture the most discriminative features among different categories of RS images. Recently, deep learning methods have gained prevalence and achieved impressive results in many research areas [13], [14], [15], [16], [29], [36], [39], [42], including image classification [13], object detection [16], and semantic segmentation [36]. In scene classification tasks, deep neural networks have been shown to be effective in extracting high-level features that can capture the complex and abstract information presented in the images. However, training a well-performing deep network from scratch requires a huge amount of data, which is undoubtedly time-consuming and laborious.

Transfer learning has emerged as a popular strategy to address this issue. It adopts models that are pretrained on large-scale datasets such as ImageNet [17] and fine-tunes them with limited RS images to achieve satisfactory performance in downstream RS scene classification tasks. Considering the significant domain gap between the characteristics of aerial images and natural images in terms of view, color, texture, layout, objects, etc., using pretrained weights from ImageNet to initialize models may limit the fine-tuning performance in downstream RS scene tasks. To overcome this limitation, Wang et al. [18], [19] retrain some classical large networks with a million-level RS dataset, which eliminated the gap between pretraining datasets and downstream datasets. With the emergence of large RS models, downstream RS tasks have

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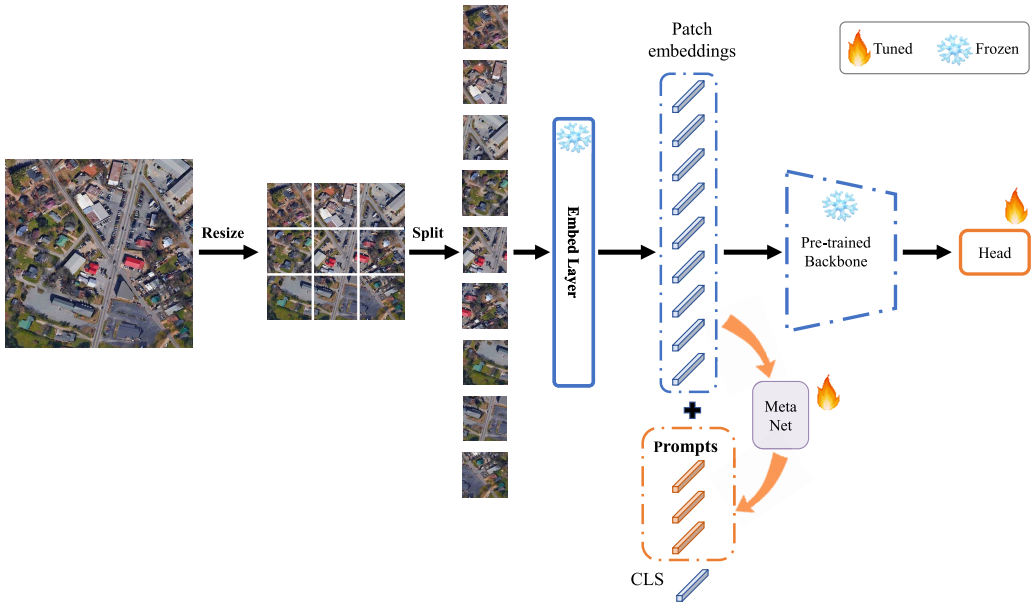


Fig. 1. Overview of IVP's framework. In the training phase, we freeze the parameters of the backbone network and adaptively generate prompts based on the input RS images. These prompts are then fed into the network along with the images and the class token (CLS) to calibrate the pretrained features.

achieved better performance and faster convergence. However, fine-tuning all the parameters of the pretrained model for each downstream task requires significant computational and storage burden, limiting its transferability to different RS classification tasks. With the continuous increase in the scale of pretrained models, this issue becomes particularly prominent [20]. Furthermore, fine-tuning a large number of weights in the pretrained model can easily lead to overfitting due to the small size of the downstream training data. Another simple solution is linear probing which is a straightforward approach to maintain the pretrained model fixed by adjusting only specific lightweight classification heads for different tasks. Although linear probing is computationally and parameter-efficient, it is generally not attractive due to its inferior performance.

In the natural language processing (NLP) domain, an effective and efficient solution is prompt learning [21], [22], [23]. The prompt is a trainable or predefined natural language sentence that is provided as additional input to a model to guide the neural network toward learning a specific task or objective [24]. Inspired by this, we propose instance-aware visual prompting (IVP), which is the first work to explore the prompting in the field of RS scene classification. Furthermore, due to the complex distribution of land cover in RS images, different categories of scenes may exhibit similar features, while scenes within the same category may also have significant differences. Learning a task-specific fixed prompt for all the images will lead to suboptimal performance. Our method adaptively generates specific prompts for each RS image and updates only a few parameters to transfer the pretrained RS model to different scene classification tasks. Our overall framework can be seen in Fig. 1. Specifically, instead of fine-tuning model parameters or adding task-specific heads, we introduce some instance-specific prompt vectors into the input space. Then, considering the significant variability in RS images, we introduce an instance-level prompt generation module, called the Meta-Net, to generate specific prompts for each RS image by aggregating contextual information from the input. Finally, these prompt vectors will calibrate the pretrained features to encode instance-specific information. To evaluate the effectiveness of our IVP method, we conduct experiments on three RS classification datasets. Our method outperforms other fine-tuning methods with only a small number of updated parameters, demonstrating the potential of our method to improve the efficiency of model tuning in RS. Especially, applying the IVP to the Swin transformer (Swin-T) backbone network outperforms full fine-tuning on all the datasets. We believe that our IVP method provides new insights for efficient model tuning and opens up new avenues for researches in RS scene classification.

The main contributions of this work are summarized as follows.

1) To the best of our knowledge, this is the first work to introduce the prompt tuning in RS scene classification, which is a parameter-efficient method for transferring large RS models to downstream classification tasks.
2) We introduce an instance-level prompt generation module, which can capture the context information of input RS data to adaptively generate specific prompt for each image.
3) The experimental results show that our method can obtain competitive performance or even better performance than full fine-tuning. For example, when updating just 1.1% parameters, the Swin-T model achieves about 1.83% and 1.42% improvement compared with the full fine-tuning method on NWPU-19 and NWPU-28, respectively.

The remainder of this article is organized as follows. Section II reviews related works. Section III introduces the proposed IVP in detail. Section IV exhibits the results from

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experiments conducted on three publicly available datasets. Finally, the conclusion of this article is presented in Section V.

II. Related Works

A. RS Scene Classification

RS scene classification is an important research topic in the RS community, and a lot of researches have been conducted for it. RS scene classification aims to accurately assign a semantic label to each given RS image. How to represent the discriminative features of the input image is the most crucial step in achieving this goal. The previous feature representations for scene classification can be roughly categorized into three types: handcrafted features, middle-level features, and deep features.

Early feature extraction methods for scene classification were based primarily on hand-crafted features which focus on texture, color, shape, or spatial information [25], such as color histogram [1], scale-invariant feature transformations [26], local binary patterns (LBPs) [27], and histograms of oriented gradients [28] or combinations of these features [30]. These methods perform well on RS images by learning regional information and discriminating the global features. However, as the scenarios become more complex and challenging, the practical application of these methods is less effective [1]. Furthermore, the middle-level feature methods [12], [31], [32] attempt to extract local descriptors of the image and then encode them to obtain a higher level feature representation. It usually encodes the obtained image descriptors with some function mapping or filtering. One of the most widely used coding methods for scene classification is the visual word bag (BoVW) [12]. Beyond that, the improved Fisher kernel [31] and the vector of locally aggregated descriptors [32] are also classical methods. These methods have been demonstrated to be effective tools for RS scene classification, but they require large amounts of prior information, and the image is represented by the number of occurrences of local features. Thus, the spatial information in the image is severely lost, which has limitations in terms of representation capability. To alleviate the difficulty of manual labeling, the unsupervised feature learning methods have gradually become popular to automatically learn features from unlabelled images [33], [34]. However, these unsupervised methods still fail to capture the most discriminative features between different categories. Further research is required to develop more effective feature representations for RS scene classification in complex and challenging scenarios.

In recent years, with the superiority of automatically extracting deep features that reflect the inherent properties of objects, deep learning has achieved an impressive breakthrough in the RS scene classification community [35], [37], [38], [40]. In the field of RS, the most commonly used deep models are the convolutional neural networks (CNNs), which have a good ability for local perception and global perception [18]. Chen et al. [41] propose the multibranch local attention network to highlight the specific channels or spatial positions of the features, which can emphasize the main target in the complex background and improve the ability of feature representation. Wang et al. [43] conduct an enhanced feature pyramid network with deep semantic embedding for classification. It improves on vanilla CNNs by extracting multiscale and multilevel features within the whole RS image. Li et al. [44] propose a fusion strategy to integrate multilayer features of the pretrained CNNs for scene classification. Compared with the previous two methods, CNN-based methods can automatically learn more class-specific and generalized features while only requiring the original input RS images without additional feature extractors, which improves the accuracy of RS scene classification.

B. Pretraining and Transfer Learning

Although the CNN-based methods in deep learning dramatically outperform previous techniques that relied on handcrafted features, training a deep CNN network from scratch in practical RS applications typically requires a large amount of high-quality labeled data [45]. However, in practical RS applications, the availability of annotated data is often limited, which is insufficient for effective training of deep learning models, leading to inferior scene classification performance. A direct solution is to use data augmentation [46] to expand the training data, or to add adversarial perturbations [47], but this requires a significant amount of computational resources. To further improve the classification performance, some methods [48], [49] have adopted transfer learning, i.e., using ImageNet pretrained models as the backbone network to use multilevel features and fine-tune parameters in the downstream datasets. Fine-tuning on specific tasks enables the rich knowledge implicitly encoded in the pretrained models to benefit a variety of downstream tasks. In fine-tuning for RS scene classification, the models use parameters trained on ImageNet and many attempts have obtained favorable results. Nevertheless, since natural images inevitably present a large domain gap relative to RS images, this may limit the fine-tuning performance of downstream scene classification tasks. Considering the gap between the natural image pretrained network and the RS tasks, Wang et al. [18] have trained a number of visual converter networks from scratch using millions of RS images and fine-tuned them for downstream scene classification tasks to achieve performance beyond that of the state-of-the-art (SOTA) methods.

Nevertheless, the existing fine-tuning methods mainly rely on updating all the network parameters, which is not an efficient learning approach. Not only does it consume a lot of computational resources but also it needs to store a set of parameters for each downstream task. To address this challenge, some researchers have proposed more efficient ways to fine-tune pretrained models on target tasks. Inspired by prompting in NLP, VPT [50] proposes to insert a small number of learnable parameters to the model's input, namely, prompt, and optimize them while freezing the backbone, which achieves significant performance improvement compared with the full fine-tuning. Through prompt guidance, the model learns to associate pretrained features with the features relevant to the downstream task. Some works also propose to insert adapters in parallel into the feedforward network (FFN) of

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the original ViT model to fine-tune in a parameter-efficient way [53]. An adapter module is typically a small nonlinear network, consisting of a linear down-projection, a nonlinear activation function, a linear up-projection, and a residual connection. In addition, there are some works in NLP to investigate which layers need to be fine-tuned. Touvron et al. [51] find that fine-tuning the weights of the attention layers and freezing the weights of the other parts is sufficient to adapt the vision transformers (ViTs) to other downstream tasks.

Compared with full fine-tuning, VPT [50] and adapter [53] are parameter-efficient fine-tuning methods that transition from directly adjusting model weights to designing adapter layers or prompts. These methods learn a shared set of parameters to extract valuable information from the entire downstream dataset and transform it into feature representations suitable for downstream tasks. While these methods have been proven to be effective, they apply the same prompts or adapter layers to all the images, overlooking the specific characteristics and content of individual images, leading to suboptimal performance. In contrast, our method adaptively generates instance-specific prompts based on the input content and context of each image or instance. This instance-specific prompt generation allows for the inclusion of more personalized and discriminative information during the fine-tuning process, thereby improving the model's ability to capture and distinguish features and categories of different images.

III. Proposed Method

We propose the IVP, a simple and effective fine-tuning method for adapting pretrained RS transformer models to downstream classification tasks. The IVP introduces learnable prompt generator and keeps the backbone frozen during the downstream training stage. In this section, we first define the notations, and then formally describe the details of IVP.

A. Preliminaries

A vanilla ViT basically consists of a patch embedding layer and several consecutively connected encoders. In a ViT, an input image $I \in R^{3\times H\times W}$ is divided and flattened into $N \times N$ nonoverlapping patches: $P = I_i | I_i \in R^{3\times h\times w}$, $1 \le i \le N$. Then, these image patches are mapped to some d-dimension vectors with positional encodings by a feature embedding module, typically a 2-D convolutional layer

$$e_i^{(0)} = \text{Embed}(I_i). \tag{1}$$

The overall combination of a prepended class token $c^{(0)} \in R^d$ and the image patches are further fed into L-layer ViT blocks for attention calculation. The whole ViT is formulated as follows:

$$\left(c^{(k)}, e_1^{(k)}, \ldots, e_N^{(k)}\right) = L_k\left(c^{(k-1)}, e_1^{(k-1)}, \ldots, e_N^{(k-1)}\right) \tag{2}$$

where $k \in 1, 2, \ldots, K$. The L_k refers to the kth transformer layer. Each layer L_k consists of multiheaded self-attention (MSA) and FFNs together with LayerNorm [52] and residual connections [13]. Finally, a neural classification head is used to map the class token of the last layer, i.e., the kth layer, $c^{(K)}$, into a predicted class probability distribution y

$$y = \text{Linear-Head}\left(c^{(K)}\right). \tag{3}$$

B. Overview of the Proposed IVP

Fig. 2 shows the concrete architecture of the proposed model. Given a pretrained ViT model, we introduce a set of M continuous embeddings of dimension d, i.e., prompts, into the input space after the embedding layer.

As mentioned above, our proposed IVP algorithm attempts to learn instance-aware visual prompts to fine-tune the pretrained RS model for better downstream task migration. Specifically, we first use a feature embedding module to embed input image patches into several context tokens as in (1). Then, the IVP uses a Meta-Net to generate sample-specific prompt tokens from the input context tokens

$$\left(p_1^{(k)}, \ldots, p_M^{(k)}\right) = \text{Meta}\left(e_1^{(k)}, \ldots, e_N^{(k)}\right) \tag{4}$$

where k represents the kth layer of the network, and M represents the number of prompts we require to generate.

The total prompt tokens and the image tokens are jointly fed into the pretrained ViT model to modify the input. During training on downstream RS classification tasks, only the parameters of the Meta-Nets and the linear head are updated while the entire transformer encoder is frozen.

C. Prompt Generators—Meta-Net

The earlier work [50] in the field of natural image processing learns shared visual prompts for each image with the pretrained model frozen. However, unlike natural images, the RS images are usually obtained by a camera in a birdview perspective lying on the planes or satellites, perceiving a wide range of land uses and land cover. Due to the diversity of geographic environments, the obtained RS images show great variability in content and also contain complex and variable background areas. In this case, assigning the shared prompts to all the images may lead to performance degradation. Therefore, to better capture the characteristics of each image instance, we introduce an additional component, called as the Meta-Net, to aggregate the image patch embeddings of each layer, which can generate the soft prompts to capture instance-specific image information.

Similar to VPT, we propose two specific implementations: IVP-shallow and IVP-deep. In the setting of IVP-shallow, Meta-Net is only used in the input space of the first transformer layer ($k \in 0$) to generate sample-specific prompt tokens. Specifically, for IVP-shallow, we first input the image tokens into a global average pooling layer to obtain a global vector, so as to adapt to the changes in data distribution

$$G^{(k)} = \text{Average Pooling}\left(e_1^{(k)}, \ldots, e_N^{(k)}\right). \tag{5}$$

Then, we build a simple feedforward layer with bottleneck architecture for representation extraction, which includes a down-projection layer with parameters W_{down} and an up-projection layer with parameters W_{up}. Assuming that the prompt length is M, we can generate an independent prompt for each instance as shown below

$$\widehat{G}^{(k)} = W_{\text{up}}\left(\text{ReLU}\left(W_{\text{down}}G^{(k)} + b_1\right)\right) + b_2 \tag{6}$$

$$\left(p_l^{(k)}, \ldots, p_M^{(k)}\right) = \text{Reshape}\left(\widehat{G}^{(k)}\right) \tag{7}$$

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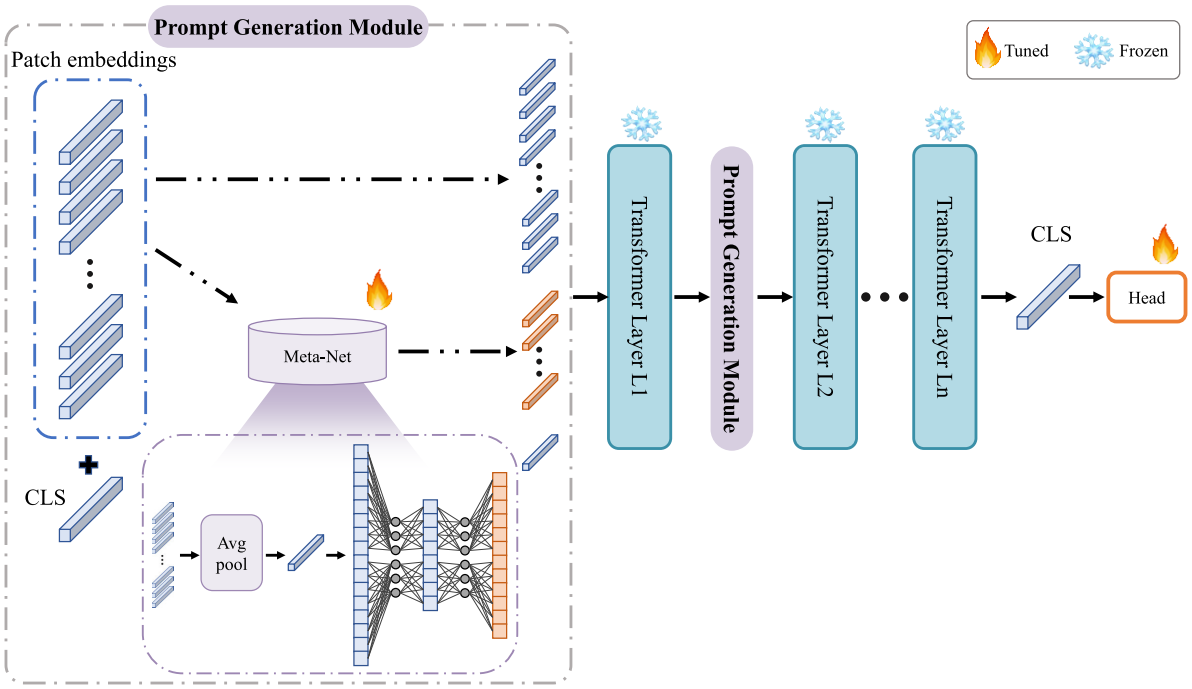


Fig. 2. Specific model architecture of IVP. We first use Meta-Net to generate sample-specific prompt tokens at each transformer encoder layer. Finally, the class token, input context tokens, and prompt tokens are concatenated as input to the frozen transformer layer. During training on downstream classification tasks, only the parameters of Meta-Net and linear head are updated, while keeping the whole transformer encoder frozen.

where $W_{\text{down}} \in R^{(d\times d')}$ and $W_{\text{up}} \in R^{(d'\times(d\times M))}$. b_1 and b_2 are the bias terms. d' is the dimension of hidden states. Finally, we have

$$\left(c^{(k+1)}, p_1^{(k+1)}, \ldots, p_M^{(k+1)}, e_1^{(k+1)}, \ldots, e_N^{(k+1)}\right) = L_k\left(c^{(k)}, p_1^{(k)}, \ldots, p_M^{(k)}, e_1^{(k)}, \ldots, e_N^{(k)}\right) \quad (8)$$

where $\{p_i^{(0)}\}_{i=1}^{M}$ is obtained from (7).

Since $d' \ll d$, the prompt generator does not have too many parameters. However, the number of parameters within W_2 will increase with the prompt length M. And for IVP-deep, each transformer layer ($k \in 0, 1, \ldots, L-1$) uses Meta-Net to generate the prompt tokens, and this number of parameters is relatively large. To reduce the computational cost, we modify the Meta-Net for IVP-deep, i.e., after the down-projection operation, we introduce an adaptive max-pooling layer to aggregate the image input tokens to generate a sequence of prompt vectors of length M, as shown below

$$\left(\hat{e}_1^{(k)}, \ldots, \hat{e}_N^{(k)}\right) = \text{ReLU}\left(W_{\text{down}}\left(e_1^{(k)}, \ldots, e_N^{(k)}\right) + b_1\right) \quad (9)$$

$$\left(\hat{p}_1^{(k)}, \ldots, \hat{p}_M^{(k)}\right) = \text{Adaptive Max Pool}\left(\hat{e}_1^{(k)}, \ldots, \hat{e}_N^{(k)}\right) \quad (10)$$

$$\left(p_I^{(k)}, \ldots, p_M^{(k)}\right) = W_{\text{up}}\left(\hat{p}_1^{(k)}, \ldots, \hat{p}_M^{(k)}\right) + b_2 \quad (11)$$

where $W_{\text{down}} \in R^{(d\times d')}$, and $W_{\text{up}} \in R^{(d'\times d)}$, and b_1 and b_2 are the bias terms. The adaptive max pool operates on the second dimension N of the input $\hat{e} \in R^{B\times N\times d'}$. It slides a variable-sized window along the input sequence and selects elements with the maximum value within each window, thereby sampling M embeddings $\hat{p} \in R^{B\times M\times d'}$. Finally, we have

$$\left(c^{(k+1)}, \tilde{p}_1^{(k+1)}, \ldots, \tilde{p}_M^{(k+1)}, e_1^{(k+1)}, \ldots, e_N^{(k+1)}\right) = L_k\left(c^{(k)}, p_1^{(k)}, \ldots, p_M^{(k)}, e_1^{(k)}, \ldots, e_N^{(k)}\right) \quad (12)$$

where $p^{(k)}$ are the prompt embeddings generated by (11), which are entered into the kth layer together with the image embeddings $c^{(k)}$. It is worth noting that the obtained outputs $\tilde{p}^{(k+1)}$ are discarded, leaving only image embeddings $c^{(k)}$. The calculations following (9)–(12) are repeated in the next layer. Specifically, in IVP-deep, a new Meta-Net is inserted in each layer. As the input sequence propagates through successive layers of the network, IVP deep introduces prompts at each layer. This accumulation of prompts at each layer leads to an increase in the length of the input sequence, which significantly increases the computational burden. Therefore, when adding new prompts to subsequent layers, we discard the prompts from the previous layer.

IV. Experimental Results

In this section, we evaluate the proposed IVP method in several downstream RS recognition tasks with pretrained RS transformer backbones. First, we describe the different benchmark datasets and the pretrained backbone that we used in these downstream tasks. Then, we briefly describe two alternative parametrically efficient transfer learning approaches and the implementation details of our experiments. Finally, we compare our method with the current benchmarks and analyze the results.

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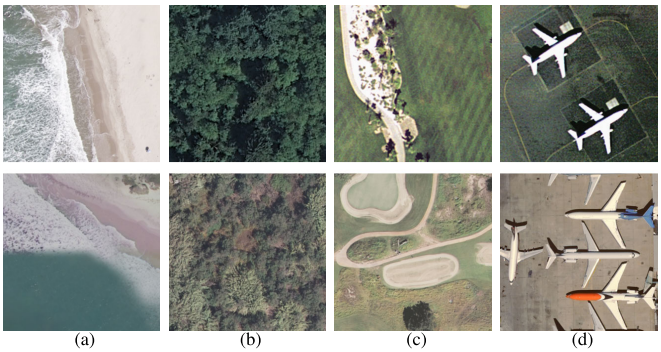


Fig. 3. Several examples of scenes in the UCM dataset. The categories are (a) beach, (b) forest, (c) golfcourse, and (d) airplane.

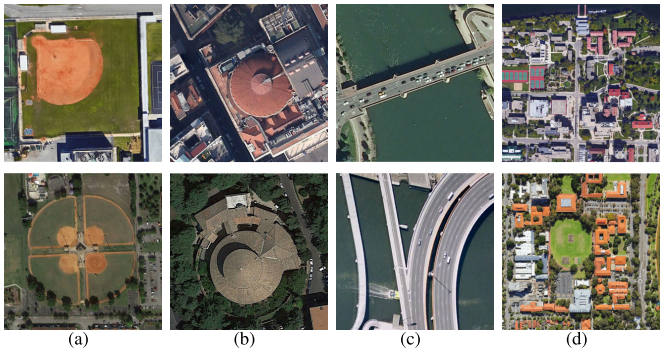


Fig. 4. Several examples of scenes in the AID dataset. The categories are (a) baseball field, (b) church, (c) bridge, and (d) school.

A. Datasets

We use the three most popular RS classification datasets to comprehensively evaluate the ability of our methods, such as the UC Merced land use (UCM) dataset [12], the aerial image dataset (AID) [2], and the NWPU-RESISC45 dataset [1] that are created by Northwestern Polytechnical University.

1) *UCM:* This is an RS dataset of land-use imagery for RS studies, with 2100 RS scene images containing three RGB channels. The images in UCM contain 21 typical land-use categories, and all the samples are extracted from the USGS National Map Urban Area Imagery series, which covers urban areas in the United States. The images in UCM have a pixel resolution of 0.3 m and an image pixel size of 256 × 256, and the categories include agriculture, airplanes, baseball diamonds, beaches, and more. Fig. 3 shows several examples of scenes in the UCM dataset.
2) *AID:* AID is a new large-scale aerial imagery dataset, which is generated by collecting images from multisource sensors on the Google Earth imager. It contains a total of 10 000 images with a pixel size of 600 × 600, covering 30 scene categories, each with approximately 220–420 aerial images. Since the image in AID is actually multisourced, it has high intraclass diversities, which brings more challenges to scene classification. Fig. 4 shows several examples of scenes in the AID.
3) *NWPU-RESISC45:* This is the available benchmark for scene classification of RS images created by Northwestern Polytechnical University, which contains a total of 31 500 images with a pixel size of 256 × 256, covering 45 scene categories, of which there are 700 images in each category. This dataset has greater intraclass differences and interclass similarities than the two above datasets, and there are also large variations in translation, spatial resolution, viewpoint, object pose, illumination, background, and occlusion. Fig. 5 shows several examples of scenes in the NWPU-RESISC45 dataset.

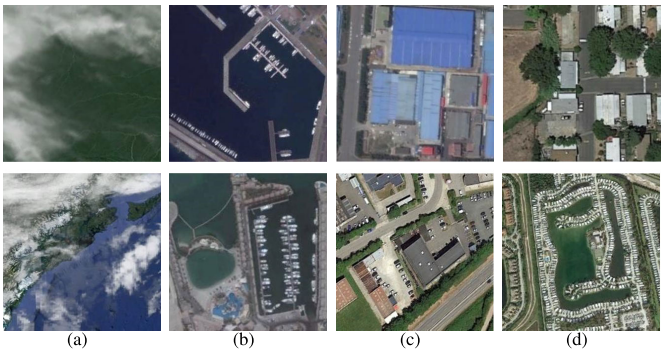


Fig. 5. Several samples of scenes in the NWPU-RESISC45 dataset. The categories are (a) cloud, (b) harbor, (c) industrial area, and (d) mobile home park.

B. Pretrained Backbones

We mainly select the large RS models pretrained on Million-AID as the backbone for fine-tuning. Following the work in [18], we conduct experiments with transformer architectures in downstream recognition tasks, including the self-supervised pretrained Swin-Ts and ViTAEv2-S (i.e., another advanced ViT, where the intrinsic biases of the CNN are introduced into the transformer), which can better capture the feature representation of RS images compared with the natural image pretrained network.

C. Baselines

First, we compare our method with the two mainstream fine-tuning methods: Full fine-tuning and linear probing. The former mainly updates all the parameters of the models, and the latter only updates the parameters of the classification head (i.e., an MLP layer). In addition, we also compare the proposed method with recent parameter-efficient fine-tuning methods: 1) adapter [53] where small modules are inserted between transformer layers and only the adapters are trained at fine-tuning time and 2) VPT [50], where the soft prompt embeddings are inserted into transformers as the input tokens and only these soft prompts are trained at fine-tuning on downstream tasks.

D. Implementation Details

For the three datasets, we follow the work in [18] settings, including UCM (8:2), AID (2:8), AID (5:5), NWPU-RESISC (1:9), and NWPU-RESISC (2:8). Note that m: n means 10 m% samples are used for training, while the others form the testing set. In particular, for the baseline methods, we follow the fine-tuning setting in [18]. Data cropping is a common

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TABLE I

PERFORMANCE COMPARISONS ON THE THREE RS BENCHMARKS WITH SWIN-T MODELS PRETRAINED ON MILLION-AID. THE BOLD FONT SHOWS THE BEST ACCURACY OF ALL METHODS AND THE UNDERLINE FONT SHOWS THE SECOND BEST ACCURACY. WE REPORT THE OVERALL ACCURACIES AND STANDARD DEVIATIONS (%)

Method \ Dataset	Average Params (M)	UCM (8:2)	AID (2:8)	AID (5:5)	NWPU-RESISC (1:9)	NWPU-RESISC (2:8)
Full fine-tuning	26.27	<u>99.53 ± 0.10</u>	<u>96.73 ± 0.07</u>	<u>97.86 ± 0.14</u>	92.03 ± 0.16	<u>93.85 ± 0.10</u>
Linear probing	0.02	97.67 ± 0.23	94.69 ± 0.06	95.55 ± 0.06	82.11 ± 0.08	83.88 ± 0.12
Adapter [53]	0.57	99.50 ± 0.18	96.49 ± 0.10	97.72 ± 0.09	90.99 ± 0.11	93.21 ± 0.21
VPT-shallow [50]	0.03	97.90 ± 0.28	95.13 ± 0.11	96.14 ± 0.07	85.65 ± 1.03	88.70 ± 1.02
VPT-deep [50]	0.23	99.29 ± 0.15	96.22 ± 0.04	97.40 ± 0.06	91.57 ± 0.16	92.98 ± 0.04
IVP-shallow (ours)	0.06	98.61 ± 0.12	95.63 ± 0.14	96.34 ± 0.14	<u>92.54 ± 0.20</u>	93.33 ± 0.08
IVP-deep (ours)	0.30	**99.76 ± 0.26**	**97.11 ± 0.13**	**97.89 ± 0.08**	**93.86 ± 0.12**	**95.27 ± 0.05**

data augmentation technique that enhances the diversity of training data. We apply cropping in our baseline approach to further improve model performance. For IVP, cropping images can lead to the loss of crucial information because it relies on extracting prompt information from the input data. Therefore, instead of using image cropping in IVP, we simply resize the input image to 224 × 224. We use the AdamW [54] optimizer to fine-tune models for 120 epochs, where the momentum is set to 0.9 and the weight decay is set to 0.05. Considering the overfitting problem, we set the basic learning rate to 6.25×10^{-5} in full fine-tuning, while in other methods, we use a larger learning rate of 3.125×10^{-3}. The overall accuracy (OA), which is the most commonly used criterion in the RS recognition community by counting the proportion of the correct classified images relative to all the images in the testing set, is used in the experiments. The models are repeatedly trained and evaluated five times in each setting, and the average value μ and standard deviation α of the results in different trials are recorded as $\mu \pm \alpha$.

For the baseline approach, we set the length of the prompts to 50 for VPT. The downward projection dimension of the adapter is 64, which is inserted at each layer of the network. Our IVP framework consists of two main versions, i.e., IVP-shallow and IVP-deep. In both the versions, we uniformly set the prompt length to 50. For the IVP-shallow, we insert the Meta-Net only in the first layer and set the downward projection dimension to 6. Note that IVP-deep introduces Meta-Net in every layer. For Swin-T and ViTAEv2-S backbone networks, their hierarchical design leads to a reduction in the resolution of the input feature map and an increase in the feature dimension at each stage. Therefore, we set the projection dimensions to the 6, 12, 24, and 48, respectively.

E. Results and Analysis

We present a comparative analysis of our IVP method with other baseline techniques on three popular RS image classification datasets. The results are shown in Tables I and II, which are evaluated in terms of Top-1 accuracy (%). The bold font shows the best accuracy of all the methods and the underlined font shows the second best accuracy.

As can be observed in Table I, our IVP outperforms other parameter-efficient fine-tuning methods, and the IVP-deep even achieves better performance than full fine-tuning in all the settings. Specifically, in the most competitive results of the two NWPU-RESISC45 dataset settings, IVP-deep obtains 1.83% (93.86% versus 92.03%) and 1.42% (95.27% versus 93.85%) accuracy improvement compared with full fine-tuning, respectively. Even with 10% RS images used for training, the simplified version of IVP-shallow achieves an accuracy improvement of 0.41% (92.54% versus 92.03%) compared with full fine-tuning. Compared with VPT, the accuracy of the shallow version of IVP improved by 6.89% and 4.63% points on NWPU-19 and NWPU-28, and the IVP-deep all received a 2.29% accuracy improvement. Moreover, the IVP-deep performs about 2.89% and 2.06% higher than adapter, and the IVP uses fewer trainable parameters.

On the AID-28, the IVP-deep still outperforms the full fine-tuning, achieving an accuracy improvement of 0.38% (97.11% versus 96.73%). On the AID-55 and ucm-82, our method also beats the full fine-tuning by a small margin. IVP-deep still achieves the highest performance among all the parameter-efficient tuning methods. However, as the amount of training data increases, parameter-efficient fine-tuning methods gradually lose their advantages over full fine-tuning in terms of accuracy. This trend is clearly reflected in the AID dataset. Comparison with the VPT method demonstrates the effectiveness of our approach—generating specific prompts for each instance yields better performance when targeting RS data. Furthermore, we outperform linear probing in all the settings.

1) IVP on Another Transformer Model: In this part, we investigate the impact of different backbone networks on the performance of our proposed IVP method. We select another Million-AID pretrained backbone network, ViTAEv2-S. The experimental results are shown in Table II. Similar to our previous results, IVP achieves superior performance over VPT and adapter, especially in the setting of "NWPU-28": IVP-deep obtains 0.93% (95.00% versus 94.07%) and 0.74% (95.00% versus 94.26%) accuracy improvement compared with IVP-deep and adapter, respectively, while our method requires only a smaller number of parameter updates. We observe that both the variants of IVP outperform linear probing, while full fine-tuning yields the highest accuracy scores overall, but at the high cost of requiring a large number of parameter updates. We believe that the introduction of the convolutional architecture in ViTAEv2-S makes it challenging to apply parameter-efficient fine-tuning methods such as IVP, since the parameters in the convolutional

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TABLE II

PERFORMANCE COMPARISONS ON THE THREE RS BENCHMARKS WITH VITAE2-S MODELS PRETRAINED ON MILLION-AID. THE BOLD FONT SHOWS THE BEST ACCURACY OF ALL METHODS AND THE UNDERLINE FONT SHOWS THE SECOND BEST ACCURACY. WE REPORT THE OVERALL ACCURACIES AND STANDARD DEVIATIONS (%)

Method \ Dataset	Average Params (M)	UCM (8:2)	AID (2:8)	AID (5:5)	NWPU-RESISC (1:9)	NWPU-RESISC (2:8)
Full fine-tuning	17.99	**99.90 ± 0.13**	**96.91 ± 0.13**	**98.22 ± 0.09**	**94.41 ± 0.11**	**95.60 ± 0.06**
Linear probing	0.01	97.00 ± 0.29	93.93 ± 0.26	94.73 ± 0.17	86.45 ± 0.27	87.07 ± 0.66
Adapter [53]	0.44	99.43 ± 0.12	95.58 ± 0.23	97.20 ± 0.17	92.34 ± 0.11	94.26 ± 0.15
VPT-shallow [50]	0.02	98.62 ± 0.18	94.46 ± 0.07	95.48 ± 0.02	88.42 ± 0.05	89.73 ± 0.07
VPT-deep [50]	0.18	99.38 ± 0.24	96.08 ± 0.12	97.21 ± 0.10	93.25 ± 0.09	94.07 ± 0.08
IVP-shallow (ours)	0.03	98.71 ± 0.15	94.97 ± 0.06	95.58 ± 0.06	89.86 ± 0.05	90.97 ± 0.04
IVP-deep (ours)	0.15	99.67 ± 0.12	96.69 ± 0.09	97.82 ± 0.10	93.58 ± 0.05	95.00 ± 0.02

TABLE III

COMPLEXITY COMPARISONS OF ADAPTER, VPT, AND OUR IVP. THE SAME PARAMETERS AND FLOPS FOR TRAINING AND INFERENCE; NO ADDITIONAL PARAMETERS AND FLOPS ARE REQUIRED FOR INFERENCE

Method	Adapter [53]	VPT-shallow [50]	VPT-deep [50]	IVP-shallow (ours)	IVP-deep (ours)
Extra Params	$2Ldd'$	Md	LMd	$dd' + Mdd'$	$2Ldd'$
Extra FLOPs	$2NLdd'$	NMd	$NLMd$	$N(dd' + Mdd')$	$2NLdd'$

architecture are not involved in the update. Nevertheless, our approach is very close to full fine-tuning, which also verifies the effectiveness of IVP on other models.

2) Complexity Analysis: We also compare the complexity of adapter, VPT and our IVP. Taking the ViT network as an example, the dimension and number of the tokens are d and N, respectively. First, assuming that the adapter architecture projects the features computed through self-attention from the d dimension to the d' dimension, then we can know that the parameter of the MLP layer is $2dd'$. On the contrary, VPT introduces additional learnable vectors into the input space. Assuming that it introduces M prompts to each layer, the additional parameter of each layer is Md. Meanwhile, our IVP is similar to the combination of VPT and adapter, where the prompt vectors are adaptively generated by an additional module, i.e., Meta-Net. For the simplified version of IVP-shallow, the parameter of the Meta-Net is $dd' + Mdd'$. However, the IVP-deep introduces a pooling operation between the down-projection and up-projection operations, which directly obtains the prompt of length M by pooling on input image tokens (i.e., pooling the input of length N to the prompt of length M). This part does not have parameters update, and the number of parameters introduced by each Meta-Net is $2dd'$. The specific number of additional parameters used by adapter, VPT, and IVP depends on the values of d, d', and M. When the total number of layers is L, the complexity of adapter, VPT, and IVP is as shown in Table III.

3) Computational Cost: To validate the efficiency of the method, we compare the training time and number of parameters of all the methods. We use a batch size of 64 and conduct a total of 120 training epochs. The experiments were performed on a single Nvidia GTX 3090 GPU, using the Swin-T model on the NWPU-28 dataset. As shown in Table IV, we can observe that the two versions of IVP require only 0.06 and 0.30 M training parameter updates, which are 0.2% and 1.1% of the full fine-tuning, respectively, but requiring more training time. Similarly, the VPT based on prompt learning also requires more training time. For further analysis, we detailed the time consumption of each process in the training phase. It is found that IVP and VPT are faster in the parameter update phase, but spend more time in the forward inference and gradient computation phases. This is due to the fact that the prompt increases the length of the network input sequence, which theoretically has a quadratic relationship with the computational cost in the multiheaded attention layer of the transformer model [50]. Furthermore, we analyzed the effect of changing the prompt length on the IVP training time of Fig. 6. In particular, when the prompt length is less than 20, the training time of IVP is significantly shorter than that of full fine-tuning, and the model performance surpasses full fine-tuning. We believe that IVP is still a parametrically efficient approach in practical applications, such as when hardware memory is limited. Our future research will focus on solving the computational cost problem of prompt learning methods to achieve a balance between performance and computational efficiency.

F. Impacts of Different Designs

We evaluate how IVP affects the fine-tuning results from several aspects, e.g., the different dataset scales, the analysis of the IVP component, and the number of prompts generated. We conduct relevant experiments to analyze the impact of different designs on the fine-tuning of backbone network. All the experiments are implemented with the pretrained Swin-T model on NWPU-RESISC45 or AID dataset, and the results are as follows.

1) IVP on Different Downstream Data Size: To explore the performance of our proposed VPT-deep algorithm on different training data scales, we conduct experiments by varying the training data from 10% to 80% and compare it with other methods. The average results for each method on different training data scales are shown in Fig. 7. The results show that IVP-deep is competitive with all other baselines

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TABLE IV

COST ANALYSIS USING A SWIN-T PRETRAINED ON SUPERVISED MILLION-AID. FOR EACH METHOD AND NWPU-28 DOWNSTREAM TASK, WE REPORT THE SPECIFIC TIME SPENT BY THE MODEL DURING TRAINING

	Total training time	Training phase breakdown time			Training parameters (M)	Accuracy (%)
		Forward Inference	Gradient computation	Parameter Update		
Full fine tuning	00:49:50	00:17:47	00:26:13	00:05:50	26.27	93.85
Linear probing	00:18:20	00:17:00	00:02:08	00:00:12	0.02	83.88
Adapter	00:35:56	00:17:18	00:17:20	00:01:18	0.57	93.21
VPT-Shallow	00:58:28	00:24:05	00:34:00	00:00:23	0.03	88.70
VPT-Deep	00:58:51	00:24:06	00:34:01	00:00:44	0.23	92.98
IVP-Shallow (ours)	00:58:20	00:24:02	00:34:00	00:00:18	0.06	93.33
IVP-Deep (ours)	01:03:33	00:24:59	00:37:38	00:00:56	0.30	95.27

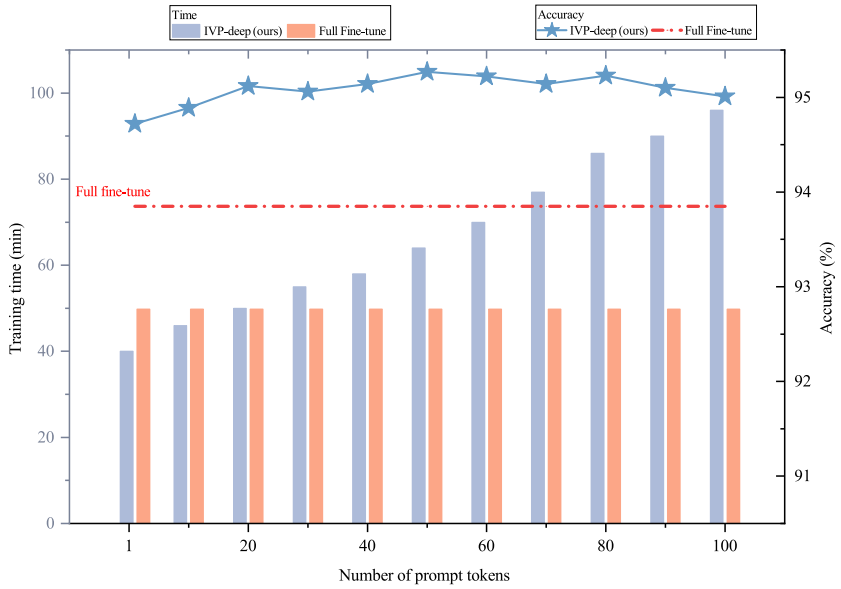


Fig. 6. Training time of IVP at different prompt lengths compared with full fine-tuning.

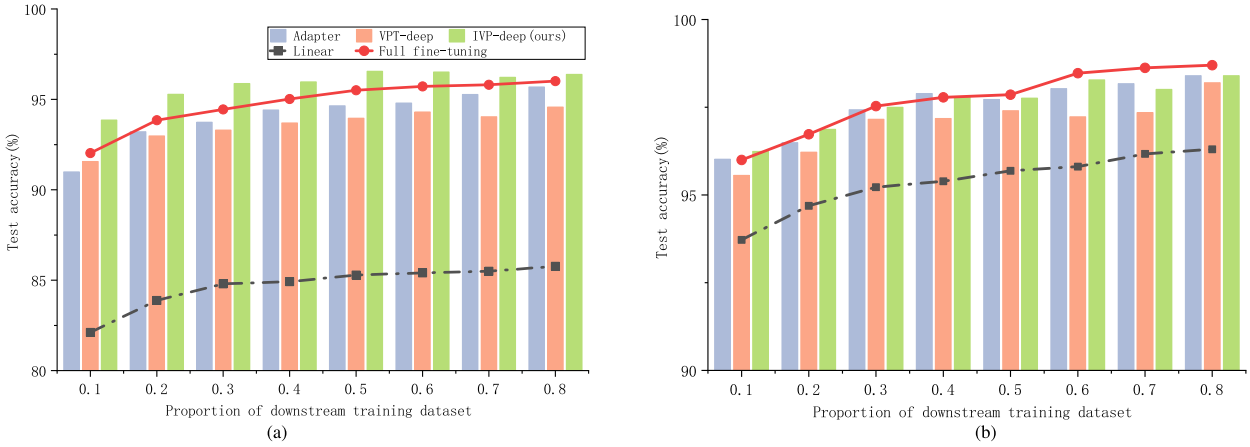


Fig. 7. Performance comparisons at different downstream data scales. IVP-deep is compared with fine-tuning, adapter, and IVP-deep. We change the holdout fraction of the training set from 0.2 to 0.8, and we show experimental results on the two benchmarks in this figure. (a) NWPU-RESISC45. (b) AID.

in terms of different data scales. Specifically, on the NWPU-RESISC45 dataset, the IVP-deep consistently outperforms full fine-tuning. Particularly, with a small amount of training data, our performance is significantly better than full fine-tuning and other methods. However, as more data are involved in training, the performance of IVP-deep gradually approaches that of full fine-tuning. On the AID dataset, our method still performs best when the training data are between 10% and 50%. In other cases, the accuracy of IVP-deep is still close to full fine-tuning and achieves better results than other methods. In summary, our method can effectively alleviate the overfitting problem of full fine-tuning with small amounts of training data and achieve encouraging results on more complex datasets. The other two parameter-efficient tuning methods can

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TABLE V

IMPACT OF THE META-NET ARCHITECTURE. OUR DATA PROCESSING STRATEGY IS ALSO APPLIED TO VPT, ELIMINATING RANDOM CROPPING AND CENTER CROPPING DURING BOTH TRAINING AND TESTING PHASES. WE REPORT THE OVERALL ACCURACIES (%)

Method \ Dataset	Average Params (M)	UCM (8:2)	AID (2:8)	AID (5:5)	NWPU-RESISC (1:9)	NWPU-RESISC (2:8)
VPT-shallow (with cropping)	0.03	97.90	95.13	96.14	85.65	88.70
VPT-shallow (without cropping)	0.03	98.33	95.59	96.32	91.66	93.09
IVP-shallow (ours)	0.03	**98.81**	**95.60**	**96.37**	**92.12**	**93.20**
VPT-deep (with cropping)	0.23	99.29	96.22	97.40	91.57	92.98
VPT-deep (without cropping)	0.23	99.35	96.79	97.38	93.76	95.01
IVP-deep (ours)	0.20	**99.51**	**96.99**	**97.80**	**93.96**	**95.11**

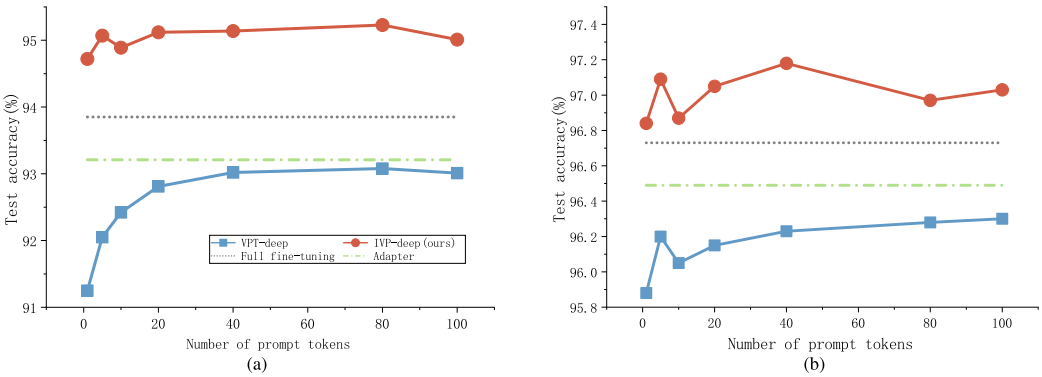


Fig. 8. Performance of VPT and IVP with different numbers of prompt tokens M. We show experimental results on the two benchmarks in this figure. (a) NWPU-28. (b) AID-28.

achieve comparable performance to full fine-tuning with small amounts of data, and as the training data increase, adapter shows a better performance than VPT-deep.

2) Analysis of the IVP Component: In the experiments mentioned above, our method is more accurate than VPT. The difference between our IVP and VPT is that the soft prompts are automatically generated by the Meta-Net based on the image instance, and there is also a difference in data preprocessing. As mentioned above, the IVP removes random cropping and center cropping during training and testing. To evaluate the effectiveness of the Meta-Net, we also remove random cropping and center cropping in the VPT, keeping the same settings as in our IVP. In addition, for a fair comparison, we reduce the parameters of the IVP to match the VPT. Specifically, we set the downward projection dimension from the original 6 to 1 in IVP-shallow, and similarly from the original 6, 12, 24, 48 to 4, 8, 16, and 32 in IVP-deep. As shown in Table V, after applying the data preprocessing pipeline of IVP, the performance of both the shallow and deep versions of VPT is significantly improved. In particular, the performance of VPT-shallow (without cropping) improved by 6.01% and 4.39% points over VPT-shallow (with cropping) on the NWPU-RESSC45 dataset. Similarly, VPT-deep (without cropping) showed an improvement of 2.19% and 2.03% points over VPT-deep (with cropping) on the same dataset. This demonstrates that an appropriate data preprocessing strategy can affect the performance of prompt learning for model fine-tuning. While cropping is removed, other stronger data augmentation strategies (Random Erasing [55], and Cutmix [56]) can be used to further improve the performance of prompt tuning. Furthermore, we can observe that the performance of both IVP-shallow and IVP-deep outperform VPT on all the datasets, and the IVP-deep can even achieve higher performance with a smaller number of training parameters. This indicates that the instance-level prompts generated by IVP can provide more effective guidance than VPT. These prompts capture specific features related to the instances, which can improve the model's performance in tasks such as RS scene classification.

3) IVP With Different Numbers of Prompt Tokens M: In this part, different numbers of prompt tokens M are used in our experiments to show the consistency of the superiority of IVP over VPT methods. Specifically, we only change the number of prompt tokens M from 1 to 100 while keeping the other experimental conditions constant. In the curves of Fig. 8(a), we can see that the IVP method achieves the highest performance when using different numbers of prompt tokens. Even when the number of prompt tokens is only one, the performance of IVP-deep is equivalent to the performance of the full fine-tuning methods. We can also see that fine-tuning works better as the number of prompt tokens increases, but it reaches saturation at a certain number. Nevertheless, the orange curve almost always lies on top of the blue curve, which means that IVP is almost always better than VPT in the test. The curve in Fig. 8(b) is unstable, with a drop in performance at some points, indicating that the appropriate number of prompts is closely related to specific downstream datasets, but in general, the optimal prompt length is around 50.

G. Visualizations

To evaluate the quality of the features produced, we conduct t-SNE [57] visualizations on IVP and other baseline methods.

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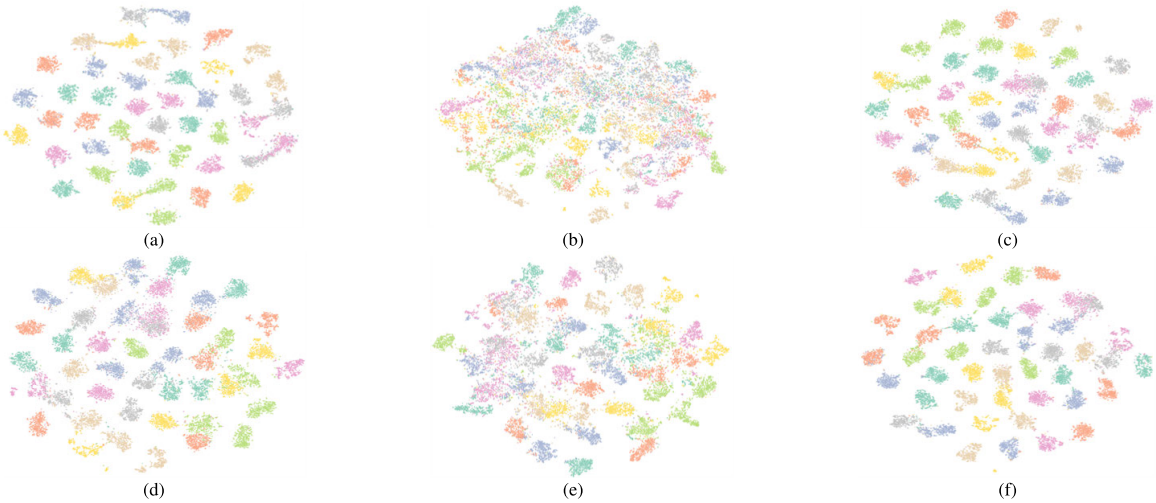


Fig. 9. t-SNE visualizations on the NWPU-28 test dataset. We extract the final classification features from the top linear layer for t-SNE visualizations. The top-1 accuracy is reported in red. (a) Full fine-tune (93.85%). (b) Linear probing (83.88%). (c) Adapter (93.21%). (d) VPT-deep (92.98%). (e) IVP-shallow (93.33%) (ours). (f) IVP-deep (95.27%) (ours).

The features are extracted from the NWPU-28 validation set via the Swin-T backbone. As shown in Fig. 9, compared with the above two parameter-efficient fine-tuning methods as VPT and adapter, the full fine-tuning strategy performs well in projecting features. However, it consumes huge computational resources to tune the entire network parameters. Although linear probing is computationally and parameter-efficient, it suffers a significant accuracy drop compared with o the full fine-tuning. All the plots show that IPT-deep enables linearly separable representations while using fewer parameters than full fine-tuning. We also observe that inserting prompts at each transformer layer (IVP-deep) improves the performance, compared with IVP-shallow, where only the prompt generation module is applied at the first layer. This confirms that our IVP facilitates the RS pretrained model to generate more separable representations with fewer learnable parameters.

V. Conclusion

In this article, we presented a new framework, IVP, which explored prompting learning in RS scene classification tasks, improving the transferability of pretrained large-scale models. Rather than fine-tuning model parameters or adding task-specific heads, we introduced prompt vectors into the model input to calibrate pretrained representations. Considering the complex characteristics of RS image changes, we further introduced an instance-level prompt generation module that captures contextual information from the input RS images to generate specific prompts for each image. Extensive experiments demonstrated the superiority of IVP over the conventional full fine-tuning methods. Moreover, by fine-tuning much fewer parameters compared with full fine-tuning, our method is GPU memory-efficient and has achieved competitive performance.

We hope that our IVP method will provide new insights into efficient model fine-tuning in the field of RS scene classification. One of our ongoing works is to explore the tasks beyond scene classification, e.g., object detection and semantic segmentation. In addition, the prompts inserted by IVP are involved in the computation of self-attention along with the images, which will extend the training time as the length of the prompts increases. We will continue to refine this in our subsequent work.

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